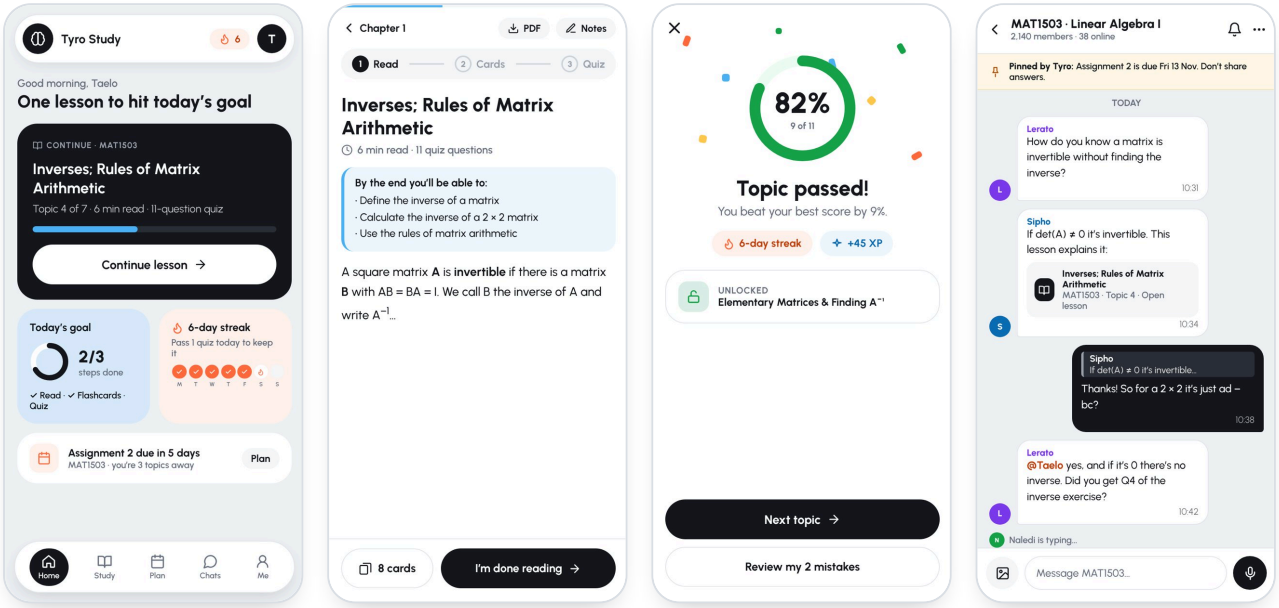
TYRO STUDY · PHASE 3 OF THE PROJECT LIFECYCLE

UX/UI design and prototyping

How students will use the redesigned Tyro Study: journeys, wireframes, high-fidelity screens, a clickable prototype, the design system, responsive and accessibility rules, and the usability test plan.

29 October 2026 · Builds on **01-discovery-and-business-analysis.md**, **02-requirements.md** and the UX review in **docs/ux-review/**



Contents

0	About this design
1	User journeys
2	Information architecture and wireframes
3	High-fidelity screens
4	Prototype
5	Design system: typography, colour and components
6	Responsive layouts and accessibility
7	Usability testing
8	Handoff and next steps

0 About this design

The latest design is the redesign from the UX review of 27 October 2026 (eight screens, drawn from the live app at www.tyrostudy.co.za), extended on 29 October with seven new phone screens and a desktop layout for the requirements added since: onboarding deadlines, module group chats, spaced-repetition review, the Me tab and the "not passed" result. Every screen uses Tyro's own font (Urbanist), colours and components.

LABEL	MEANING
UX review 27 Oct	Screen from the UX review redesign (docs/ux-review/mockups)
New 29 Oct	New screen in this document, same style
Built	Already in the app

Design principles for an audience of mostly adult distance learners, often on a phone and in their second or third language:

PRINCIPLE	WHAT IT MEANS ON SCREEN
One clear next step	Each screen has one main black button. Home always says what to do next.
Reward study, not visits	Streaks, rings and celebrations respond to lessons and passed quizzes.
Teach, don't just test	Every answer is explained; mistakes link back to the lesson.
Calm and respectful	Quiet fun (no childish mascots), no shaming copy, kind fail screens.
Thumb-first	Main actions are pinned to the bottom of the screen and at least 44 px tall.
Plain English	Short sentences, South African spelling, no idioms.

1 User journeys

Three journeys cover the moments that decide whether students stay: the first session (activation), the daily habit (retention) and getting help (the new module groups). Feelings come from the UX review walkthrough and should be confirmed in usability testing (section 7).

Journey 1 · First visit to first pass

Who: Working UNISA student, studies on a phone after work · **Goal:** find help for a module and know it works, in one sitting

	Discover	Sign up	Set up	Learn	Prove it
DOING	Googles a topic, lands on a free note	Email, password, level, 6-digit code	Picks modules and a deadline	Reads the first lesson	Takes the topic quiz
SCREEN	Public note	Register	Onboarding 2–3	Lesson	Quiz, feedback, results
FEELING	Curious	Impatient	Hopeful	Focused	Proud
PAIN TODAY	Note is good but the next step is a sign-up wall	Code email may land in spam	Today: empty dashboard, join modules one by one	Long scroll, quiz button at the very end	Submit hidden on phones; no reason when wrong
DESIGN RESPONSE	Sample quiz on the note, then "Create a free account"	Clear "check spam / resend in 60 s" copy	One multi-select step, deadline and daily goal	Read → Cards → Quiz steps, sticky "I'm done reading"	Focus mode, explanation per answer, celebration

Journey 2 · Daily 10-minute study

Who: Same student, two weeks in, on the taxi home · **Goal:** keep the streak and move one topic forward

	Prompt	Resume	Revise	Practise	Close
DOING	Gets a 19:00 reminder	Opens the app	Reviews due flashcards	Quiz, misses two	Sees streak kept
SCREEN	Push notification	Home	Review cards	Quiz → Not passed	Results / Home
FEELING	Neutral	Oriented	Efficient	Frustrated	Satisfied
PAIN TODAY	Bell dot looks like an alert all the time	"Up next" can point at a locked topic	Flashcards only exist inside a topic tab	Same screen for pass and fail	Streak counts visits, not study
DESIGN RESPONSE	One meaningful reminder a day, at the chosen time	Big Continue card, always an unlocked topic	Daily due cards with spaced repetition	"So close" screen, review mistakes first	Streak counts a lesson or a passed quiz

Journey 3 · Getting unstuck in a module group

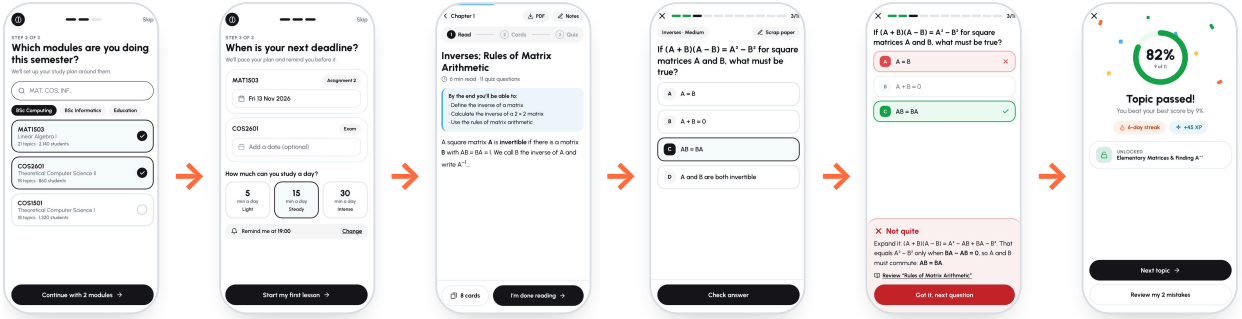
Who: Student stuck on an exercise late at night · **Goal:** get a quick, trustworthy answer from classmates

	Stuck	Ask	Answer	Notified	Safe
DOING	Can't see why a matrix isn't invertible	Opens the MAT1503 group, asks	Classmate replies with a lesson link	Gets a mention later	Sees answer-sharing, reports it
SCREEN	Lesson / quiz	Module groups → thread	Thread, topic card	Push (mention)	Message actions
FEELING	Stuck	Hopeful	Relieved	Included	Reassured
PAIN TODAY	Only 1:1 chat with friends they already know	Has to find a friend first	Links are plain text	Risk of noisy groups	No report or block today
DESIGN RESPONSE	Every module has its own group	Group joined automatically with the module	Tyro links show as cards that open the lesson	Mentions and replies only by default	Report, block, admin moderation, pinned rules

Personas behind the journeys

PERSONA	CONTEXT	NEEDS FROM THE DESIGN
Working distance learner (core)	UNISA, part-time, mid-range Android, prepaid data, studies in short sessions	Fast resume, small data use, clear plan, explanations in plain English
Matric learner	Grade 11–12, phone-only, revises before exams	Same flows with "subjects" and "grades"; safe, level-only groups
Content admin	Builds modules from textbooks with AI, desktop	Fast editing, publish and lock controls, moderation queue, analytics

Journey 1 at a glance



2 Information architecture and wireframes

The bottom navigation becomes five short labels. **Logout moves into Me** so it can't be tapped by mistake, and **Chats becomes module groups**.

Home

Continue card
Today's goal
Streak (week view)
Next deadline

Study

My modules (+ Add)
Module → chapter path
Lesson · Cards · Quiz
Chapter quiz · Mock exam
Review cards (due)

Plan

Timetable
Study sessions
Assignment and exam dates
Reminders

Chats

One group per joined module
Thread · pinned notice
Report · block
Notification setting

Me

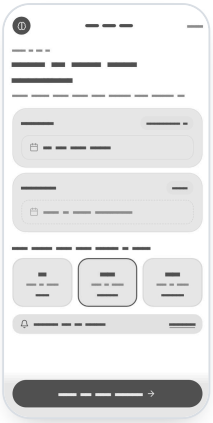
Profile · level
Daily goal · notifications
Privacy and data
Sign out
Admin (admins only)

Wireframes

Low-fidelity wireframes show layout and hierarchy without colour or copy, so structure can be judged on its own. They are generated from the same screens, so they always match the high-fidelity design.



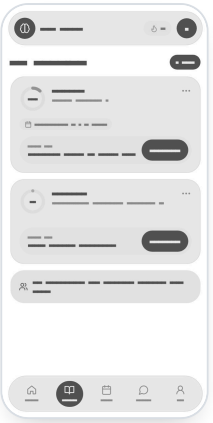
Onboarding · modules



Onboarding · deadline



Home



My modules



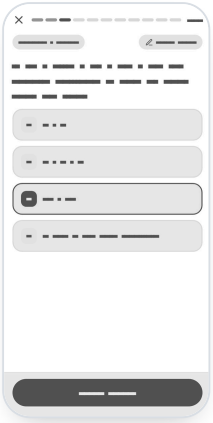
Chapter path



Lesson



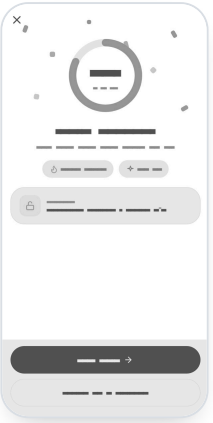
Review cards



Quiz



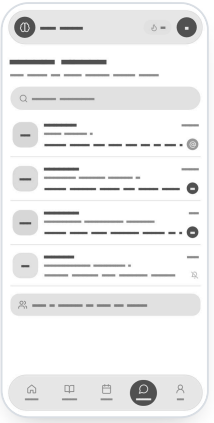
Feedback



Passed



Not passed



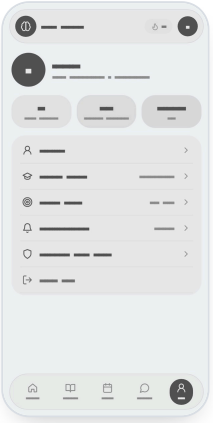
Module groups



Group thread



Message actions

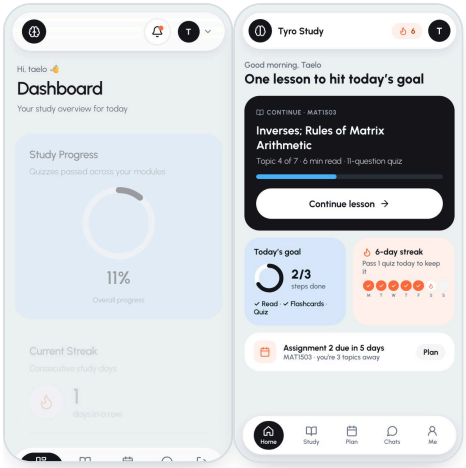


Me

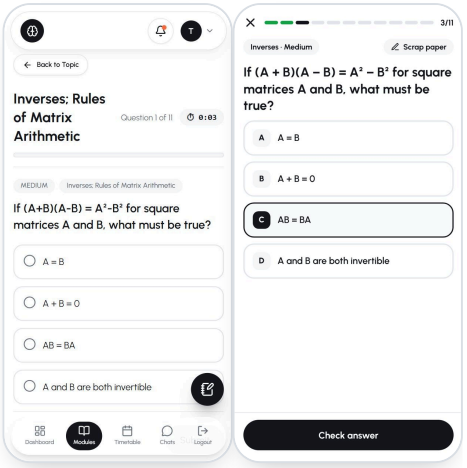
3 High-fidelity screens

Phone size 390 × 780. Notes explain the decisions; requirement ids refer to 02-requirements.md.

Before and after



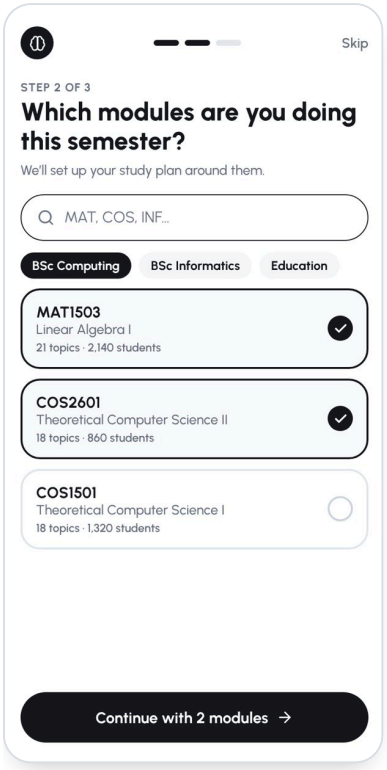
Home now | Home redesigned



Quiz now | Quiz redesigned

Now: stats first, the next step third and small; on phones the quiz Submit button sits under the nav bar next to Logout. Redesigned: the next step is the first thing you see, and quizzes open in focus mode.

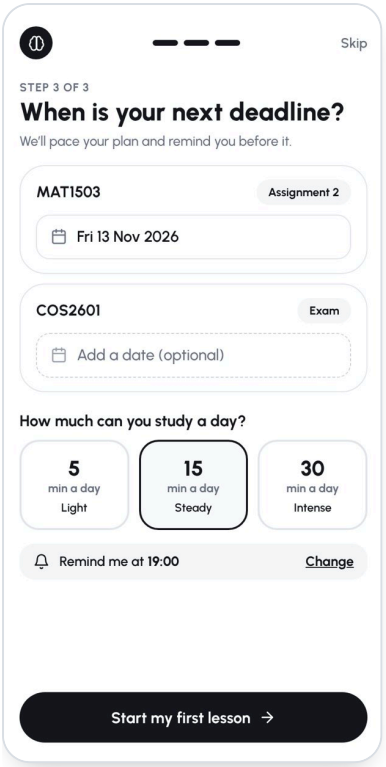
Getting started



Onboarding · pick modules UX review 27 Oct

FR-12, FR-13

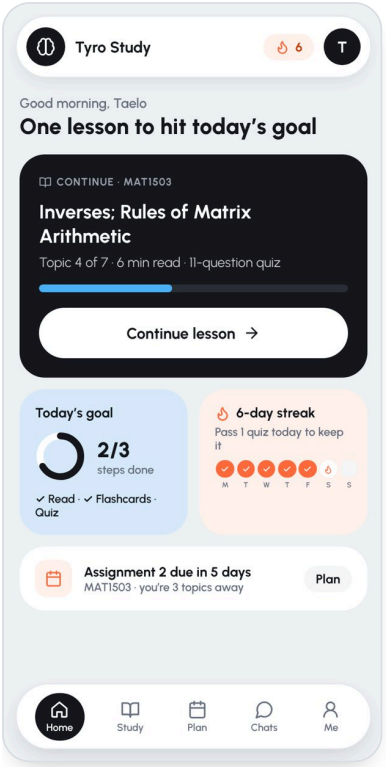
- Progress dots and Skip, so nobody feels trapped.
- Search by code and course chips; tick several modules, then one Continue.
- Student counts (from enrolments) as social proof.



Onboarding · deadlines and daily goal New 29 Oct

FR-12, FR-52, FR-53

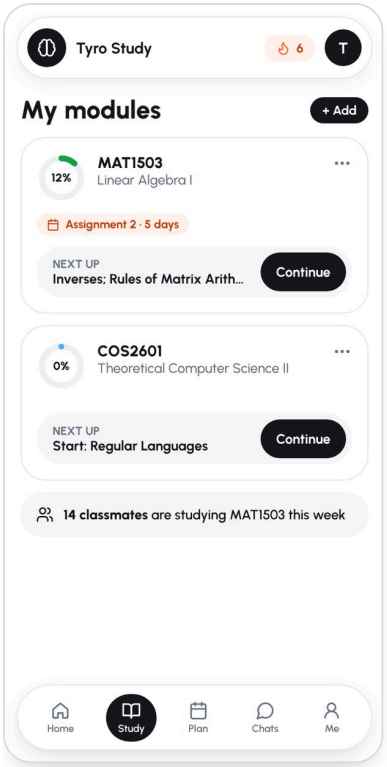
- Optional deadline per module feeds reminders and the Home nudge.
- 5 / 15 / 30 minutes sets the size of the daily goal.
- Reminder time asked here, not with a red bell dot.
- Ends with "Start my first lesson": aim for a passed quiz within 10 minutes.



Home UX review 27 Oct

FR-23, FR-51, FR-53

- Headline says what to do today.
- Black Continue card: the first unlocked, unpassed topic (fixes the locked "Up next" bug).
- Daily goal ring and a streak that shows the week.
- One deadline nudge tied to progress. New bottom nav.

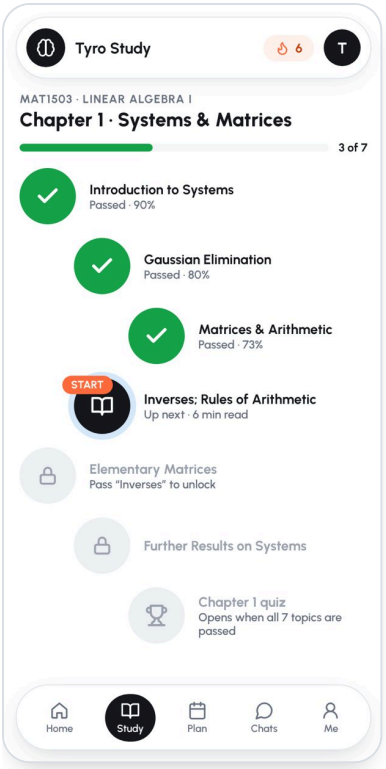


My modules UX review 27 Oct

FR-11, FR-50

- Progress ring and Next up on every card.
- Leave moves into the ⋯ menu.
- + Add replaces the separate Browse tab.

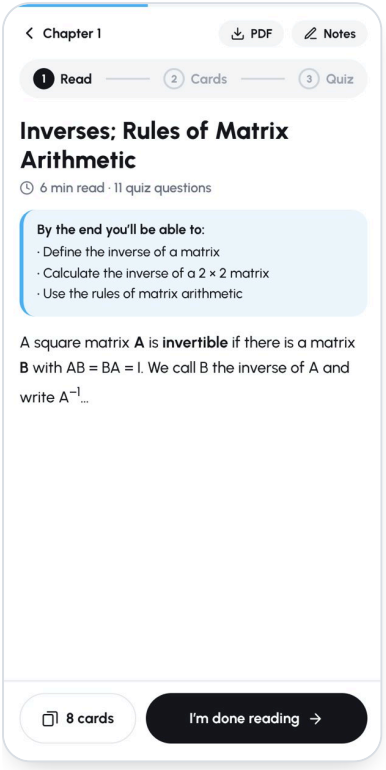
Studying



Chapter path UX review 27 Oct

FR-54, BR-20

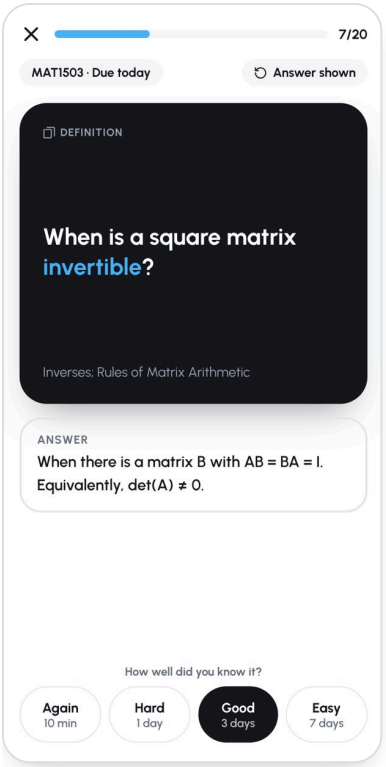
- Chapter shown as a path: passed (✓ and score), one current node, locked "coming next".
- Lock reason in words: "Pass 'Inverses' to unlock".
- Chapter quiz is the trophy at the end.



Lesson UX review 27 Oct

FR-20, FR-24

- Reading progress bar and Read → Cards → Quiz steps.
- Reading time, quiz size and learning outcomes up front.
- Sticky bottom bar: cards and "I'm done reading".

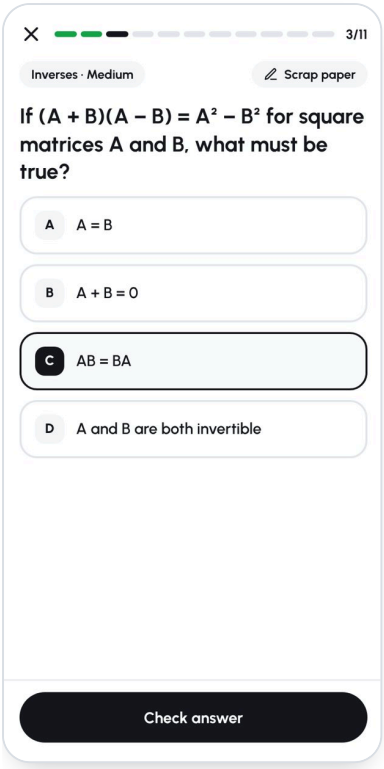


Review cards (spaced repetition) New 29 Oct

FR-41

- A short daily review of cards that are due.
- Again / Hard / Good / Easy schedules the next review (10 min to 7 days).
- Counts towards the daily goal, so five minutes still keeps the streak.

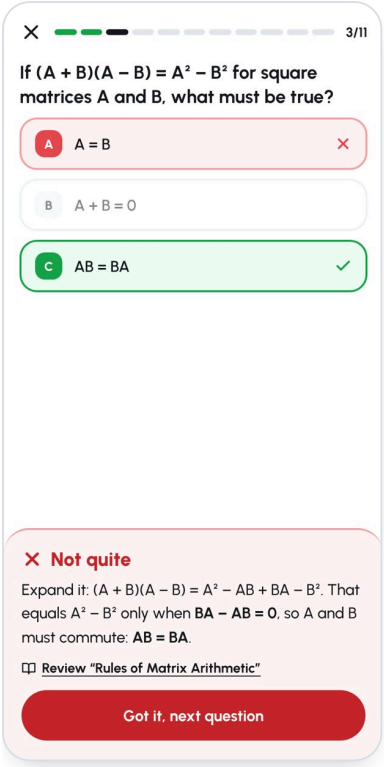
Quizzes



Quiz · focus mode UX review 27 Oct

FR-31, FR-35, FR-36

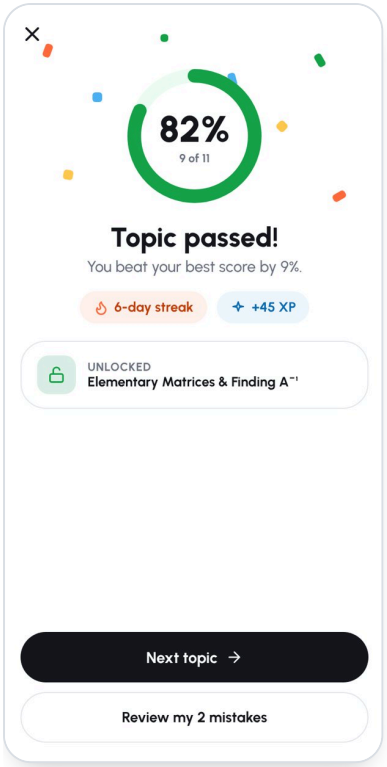
- No bottom nav; ✕ exits and keeps answers.
- Segmented progress shows right and wrong so far.
- Scrap paper in the header; sticky Check answer; A–D letters.



Answer feedback UX review 27 Oct

FR-32

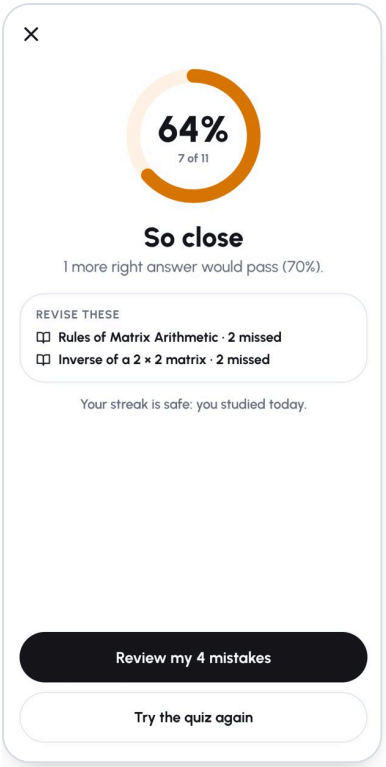
- "Not quite" with icon and colour (never colour alone).
- Two-line explanation of the reasoning.
- Link to the right part of the lesson, then one big button.



Quiz passed UX review 27 Oct

FR-34

- Celebration, score ring, gain over best score.
- Shows what it unlocked, so locks feel like progress.
- One main button (Next topic), one secondary.

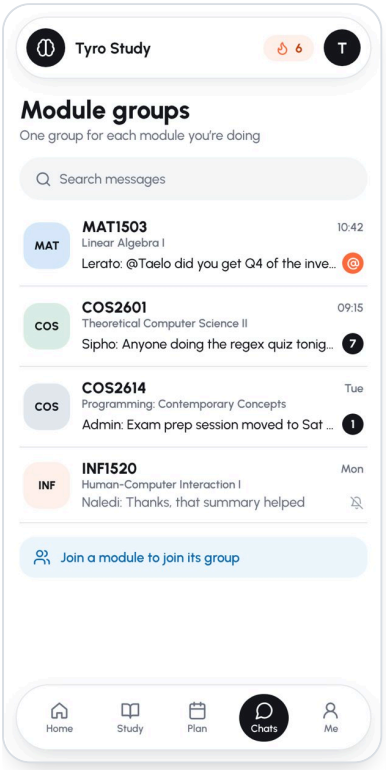


Quiz not passed New 29 Oct

FR-34, BR-20

- Kind and specific: "1 more right answer would pass".
- Lists the topics to revise.
- Review mistakes is the main action; trying again is second.
- Reassures that the streak is safe (study counted).

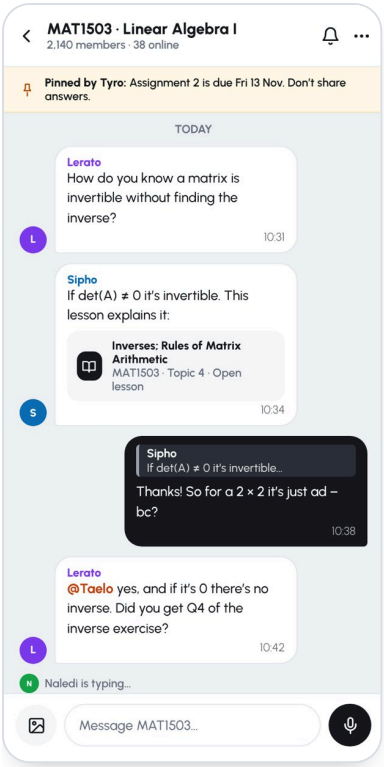
Module group chats



Module groups New 29 Oct

FR-63, FR-64, FR-66

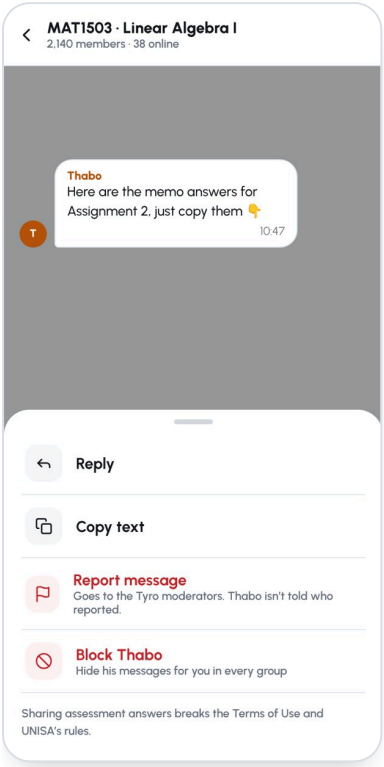
- One row per joined module, newest activity first.
- Orange @ badge for mentions; counts for unread; muted groups dimmed.
- Empty state explains that joining a module joins its group.



Group thread New 29 Oct

FR-65, FR-6A, FR-69

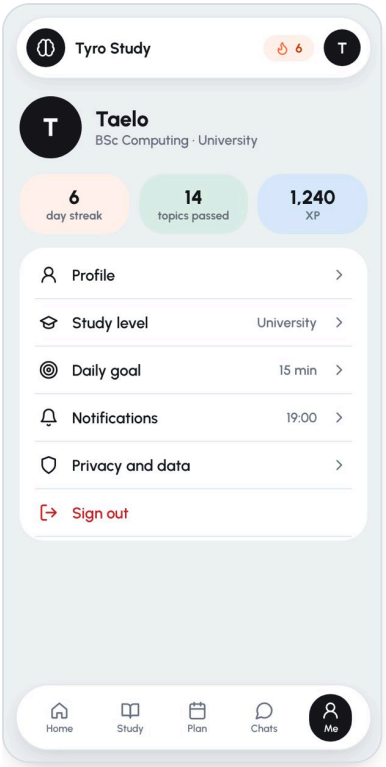
- Header shows members and an online count (from last-seen, not Pusher presence, which caps at 100).
- Pinned notice from Tyro with the no-answer-sharing rule.
- Replies, @mentions and Tyro links shown as lesson cards.
- Image and voice note buttons as in today's chat.



Message actions New 29 Oct

FR-68, BR-75

- Long-press a message: Reply, Copy, Report, Block.
- Report is anonymous and goes to the admin moderation list.
- Block hides that person's messages for you in every group.
- Short reminder of the academic integrity rule.



Me New 29 Oct

FR-05, FR-06

- Profile, level, daily goal, notifications, privacy.
- Sign out lives here, away from the main navigation.
- Small stats row: streak, topics passed, XP.

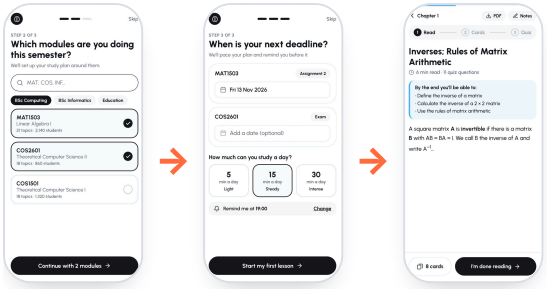
4 Prototype

A clickable prototype of all 15 phone screens is saved next to this document as **business/prototype/index.html**. Open it in any browser: tap where you would in the app, use Back and Restart, and switch on *Show hotspots* to see what is tappable. It works offline and on a phone, so it can be used for usability testing straight away.

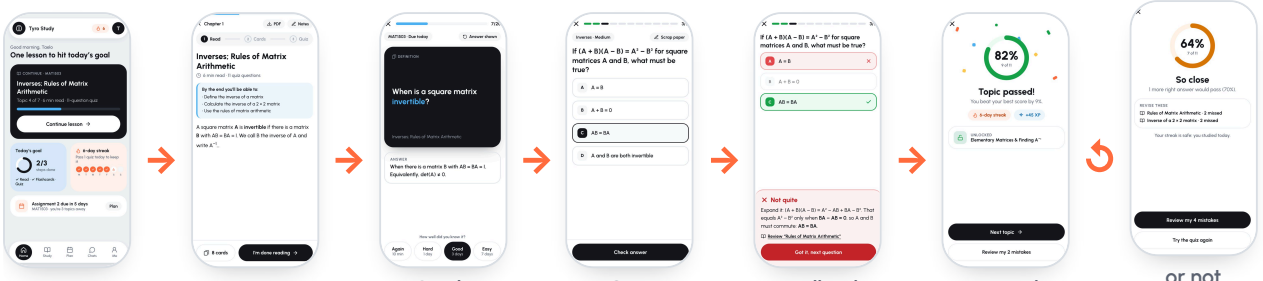
> The prototype is screen-to-screen only: no typing, animation or real data. That is enough to test whether people find the next step and understand each screen, which is what the first round of testing is for.

Flows wired in the prototype

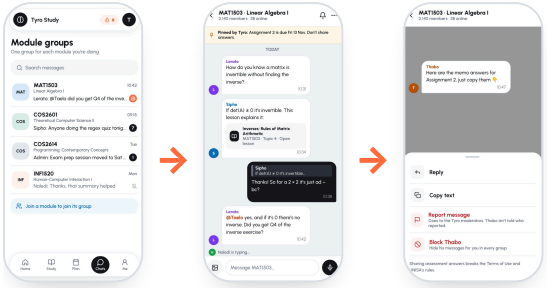
ONBOARDING



DAILY STUDY LOOP



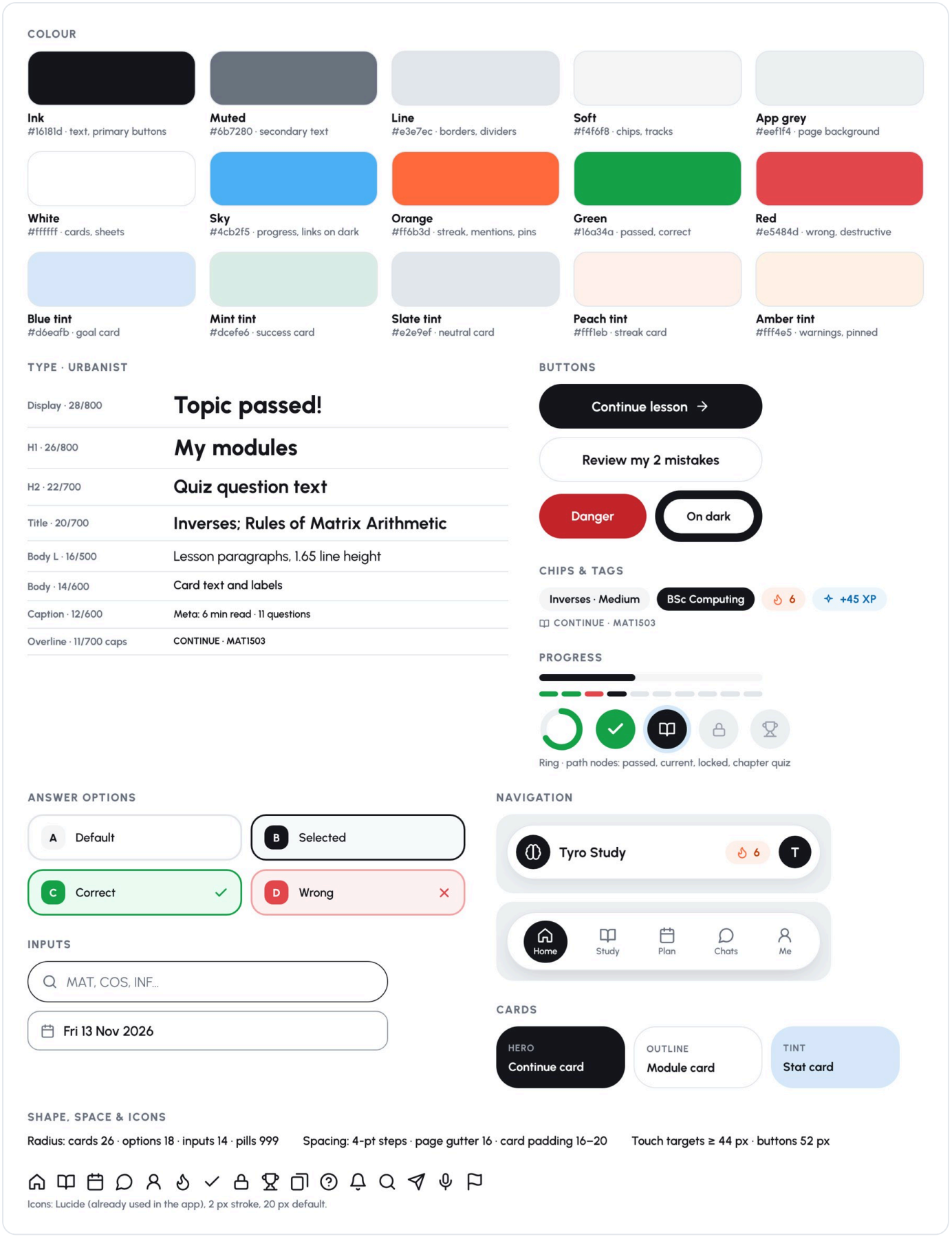
MODULE GROUPS



FROM	TAP	GOES TO
Home	Continue lesson · deadline card · Study / Chats / Me	Lesson · My modules · tabs
My modules	Continue on MAT1503	Chapter path
Lesson	8 cards · I'm done reading · back	Review cards · Quiz · Chapter path
Quiz, feedback, results · groups	Check answer · Got it · Next topic · MAT1503 row · a message	Next screen in the flow

5 Design system

The system is already in the code as CSS variables (**--neo-*** in src/app/globals.css) and in the mockups (**base.css**). This page is the reference for both, so new screens and components stay consistent.



Tokens

TOKEN	VALUE	USE
--neo-text / ink	#16181d	Text, primary buttons, current node
--neo-muted	#6b7280	Secondary text on white
--neo-border	#e3e7ec	Dividers, card outlines
--neo-surface-soft	#f4f6f8	Chips, bar tracks
--neo-accent	#4cb2f5	Progress, highlights on dark
--neo-orange	#ff6b3d	Streak, mentions (icons)
--neo-success	#16a34a	Correct, passed
--neo-danger	#e5484d	Wrong, destructive
--neo-tint-blue / mint / slate	#d6eafb · #dcefe6 · #e2e9ef	Stat and info cards

TOKEN	VALUE
Font	Urbanist 400–800 (next/font, self-hosted)
Type scale	28 · 26 · 22 · 20 · 16 · 14 · 12 · 11 px
Line height	1.15 headings · 1.45 UI · 1.65 lessons
Radius	Cards 26 · options 18 · inputs 14 · pills 999
Spacing	4-pt steps; gutter 16; card padding 16–20
Elevation	Bars: 0 8px 24px –8px rgba(22,24,29,.2); cards flat
Heights	Buttons 52 · inputs 46–48 · nav 68 · top bar 64
Icons	Lucide, 2 px stroke, 20 px
Motion	150–250 ms ease-out; respect prefers-reduced-motion

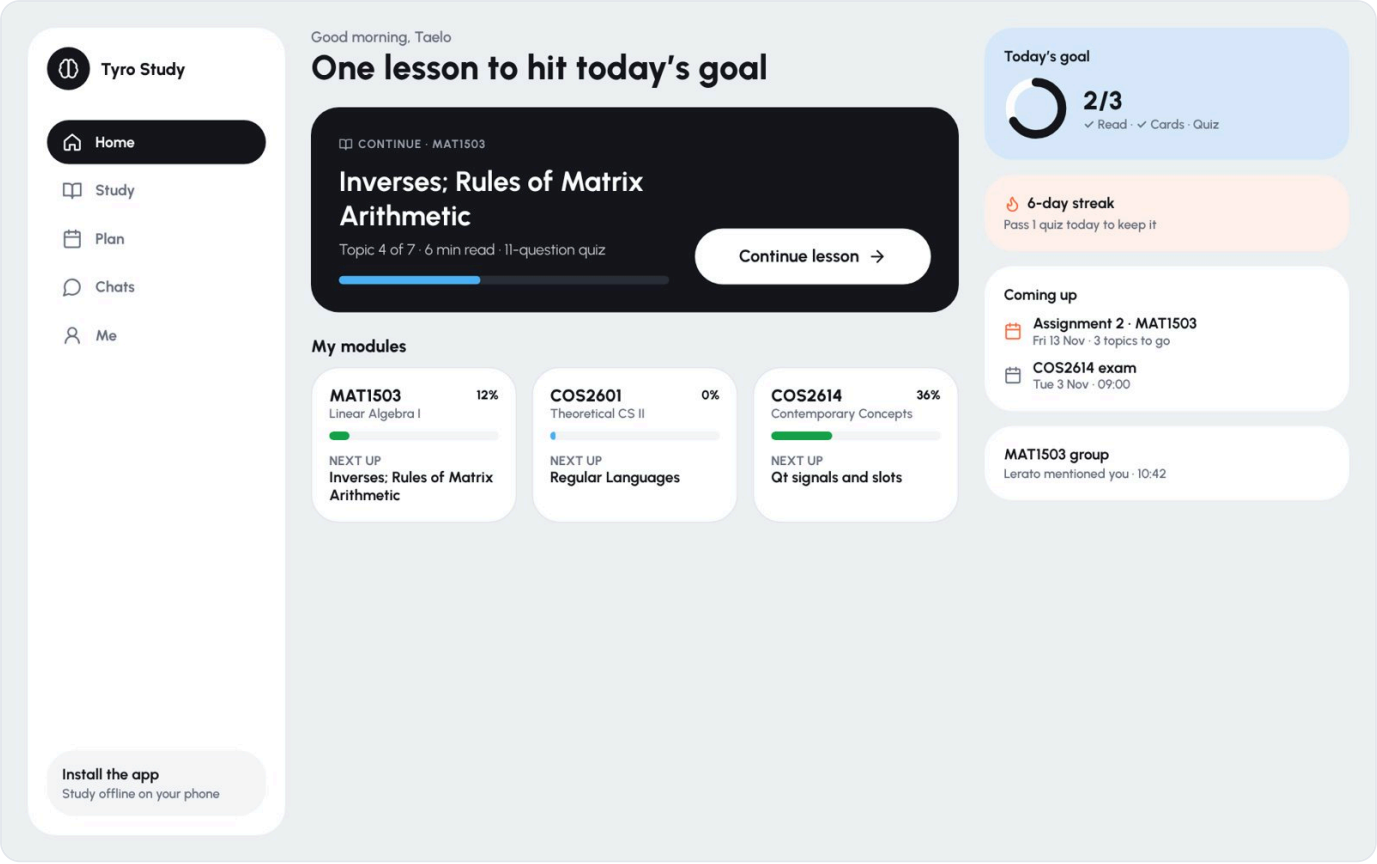
Components

COMPONENT	VARIANTS AND STATES	IN CODE
Button	Primary (ink) · white on dark · ghost · danger; disabled; loading	components/ui/button.tsx
Card	Hero (ink) · outline · tint · flat (neo-flat)	components/ui/card.tsx
Chip / tag	Neutral · selected · streak · XP · overline tag	components/ui/badge.tsx
Answer option	Default · selected · correct · wrong · dimmed	quiz-runner.tsx (extract to ui/)
Progress	Bar · segmented quiz bar · ring · path node (4 states)	New: ui/progress.tsx, ui/path-node.tsx
Navigation	Top bar (streak chip, avatar) · bottom nav (5 items) · desktop sidebar	top-nav.tsx, mobile-nav.tsx
Sheet	Bottom sheet: feedback, message actions, confirmations	New: ui/sheet.tsx (replaces alert/confirm)
Message bubble	Mine · theirs · reply quote · link card · removed by moderator	chat/message-thread.tsx
Input	Search pill · field with icon · date; focus ring; error text	components/ui/input.tsx
Toast · skeleton	Success · error; skeleton cards instead of "Loading…"	ui/toast.tsx, ui/skeleton.tsx

6 Responsive layouts and accessibility

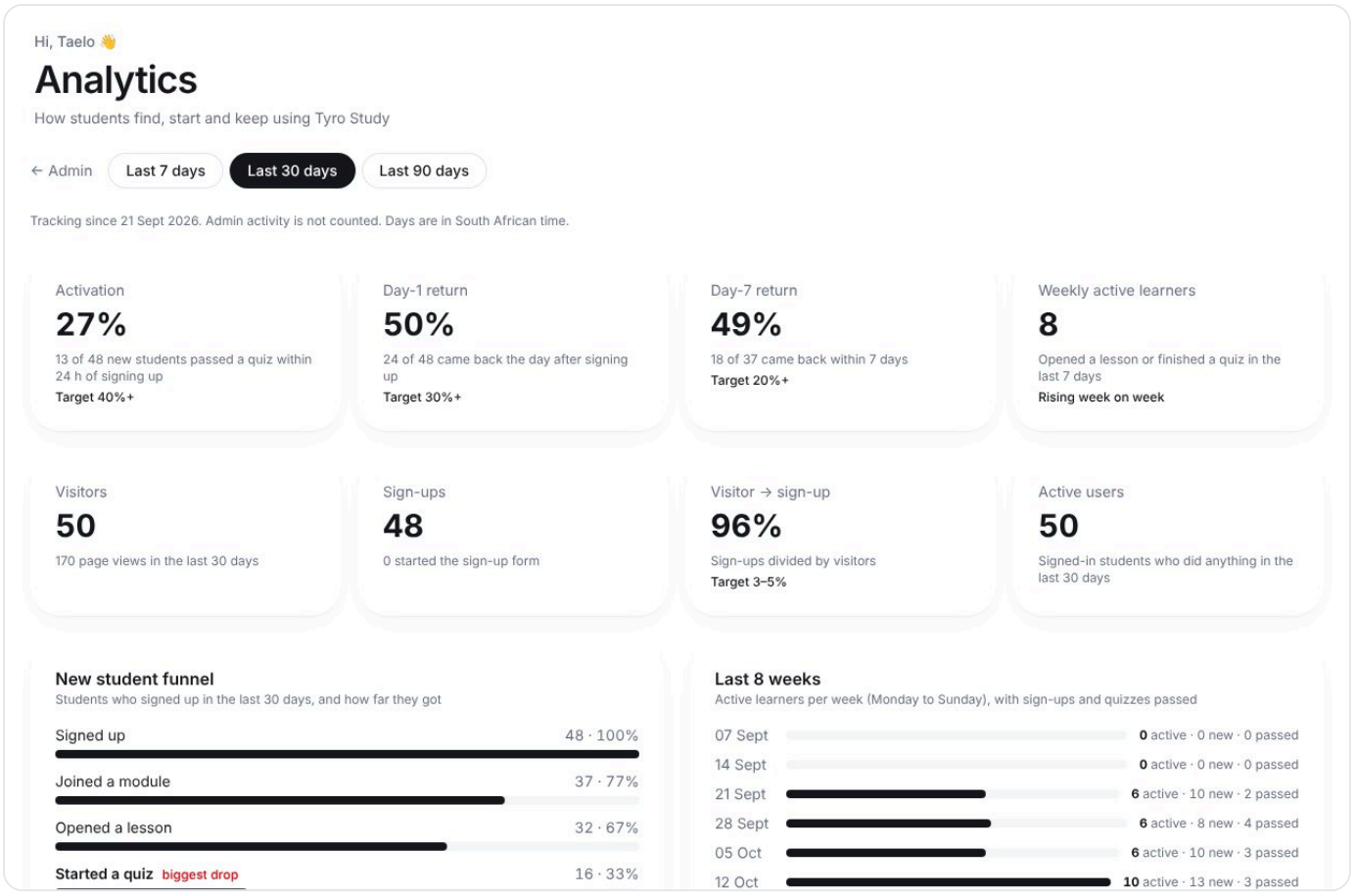
Breakpoints

WIDTH	LAYOUT	NAVIGATION
360–639 px (phones)	One column, 16 px gutters, sticky bottom actions	Floating bottom nav; hidden in quizzes, lessons and threads
640–1023 px (tablets)	Two-column cards; lesson text max 680 px	Bottom nav or top nav
≥ 1024 px (desktop)	Sidebar · main · side panel (goal, deadlines, group mentions)	Left sidebar; desktop notepad stays



Home at 1280 px **New 29 Oct**: the same Continue card and goal, with My modules and Coming up beside it instead of below.

Admin analytics (built, sample data)



/admin/analytics, added 29 October 2026 **Built**. Shown with test data. On phones the metric cards go two across and tables scroll inside their card.

Colour contrast (WCAG 2.2 AA)

Worked out from the actual token values. Text needs 4.5:1 (3:1 for large text); icons and field borders need 3:1.

PAIR	COLOURS	RATIO	WHERE	RESULT	FIX
Ink text on white	#16181d on #ffffff	17.76:1	Body text	**Pass**	–
Muted text on white	#6b7280 on #ffffff	4.83:1	Secondary text	**Pass**	–
Muted text on app grey	#6b7280 on #eef1f4	4.26:1	Captions on grey pages	**Fail**	Use #4b5563 (6.67:1) for text on grey
Locked / disabled grey	#9ca3af on #ffffff	2.54:1	Locked topic titles	**Fail**	Use #6b7280 + lock icon + "Pass X to unlock"
White on orange	#ffffff on #ff6b3d	2.83:1	START chip, @ badge	**Fail**	White on #c2410c (5.18:1); keep #ff6b3d for icons only
White on green	#ffffff on #16a34a	3.30:1	Passed node (icon only)	**Pass**	Fine for icons; use #15803d (5.02:1) if text
Dark orange on peach	#c2410c on #fff1eb	4.69:1	Streak chip	**Pass**	–
Error red on pink	#c4262b on #fff4f4	5.33:1	"Not quite" sheet	**Pass**	–
Blue on light blue	#0b6db3 on #eef7fe	5.03:1	Info notes, XP chip	**Pass**	–
Grey on ink	#9ca3af on #16181d	6.99:1	Captions in hero card	**Pass**	–
Input border on white	#e3e7ec on #ffffff	1.24:1	Text fields (non-text)	**Fail**	Field borders #8a929c (3.15:1)
Amber ring on amber tint	#d97706 on #fff4e5	2.93:1	Score ring (non-text)	**Fail**	Add the score as text (done) and use #b45309

Accessibility checklist

RULE	STATUS AND ACTION
Pinch-zoom allowed (WCAG 1.4.4)	**Fails today: remove *maximumScale: 1* and *userScalable: false* from src/app/layout.tsx (NFR-40)**
Never colour alone for right/wrong, locked, passed	Designed with icons and words on every state
Touch targets ≥ 44 × 44 px; nothing under fixed bars	Buttons 52 px; bottom nav hidden in focus screens
Keyboard and screen reader	Options are radio buttons in a fieldset; sheets trap focus and close with Esc; visible 2 px focus ring in ink
Announce results	"Correct" / "Not quite" in an aria-live region
Reduced motion	Confetti and ring animations off when prefers-reduced-motion is set
Readable lessons	16 px body, 1.65 line height, max 70 characters a line, maths rendered as text not images
Plain language	Short sentences; consistent words ("module" / "subject" by level)
Text alternatives	Alt text for lesson images; transcripts for voice notes (later)

7 Usability testing

> No usability test has been run on the redesign yet. This is the plan for the first round, using the clickable prototype, then a second round on the built screens.

Round 1 · prototype (before building)

Participants	6 UNISA students (mix of first- and second-year, at least 3 working part-time, at least 2 who study in their second language) plus 2 Grade 12 learners. Recruit from classmates and module WhatsApp groups.
Format	30 minutes, remote (video call, screen share) or in person on the participant's own phone. Think aloud. One facilitator, one note-taker.
Incentive	Mobile data voucher (R50–R100).
Consent	Short POPIA consent: what is recorded, kept for 90 days, used only to improve Tyro.

Tasks

#	TASK (READ TO PARTICIPANT)	SUCCESS MEANS	JOURNEY
1	"You've just signed up. Set Tyro up for this semester."	Picks 2 modules and a deadline, reaches the first lesson	J1
2	"You have 10 minutes on the taxi. Carry on studying."	Taps Continue on Home without help	J2
3	"Finish reading and test yourself."	Finds "I'm done reading", then the quiz	J1
4	"You got this question wrong. Why?"	Explains the reason from the feedback sheet	J1
5	"What happens next after this result?"	Understands what was unlocked / what to revise	J1, J2
6	"Ask your MAT1503 classmates for help."	Opens Chats → MAT1503 group	J3
7	"Someone is posting assignment answers. Deal with it."	Long-presses and reports	J3
8	"Sign out."	Finds Sign out in Me	–

What we measure

MEASURE	HOW	TARGET
Task success	Completed without help	≥ 80% per task
Ease per task	Single Ease Question (1–7) after each task	≥ 5.5 average
Overall usability	System Usability Scale (10 questions) at the end	≥ 75
Wording problems	Words participants misread or ask about	Fix every one seen twice
Wrong taps	Taps on non-hotspots (note-taker)	Find the confusing spots

After launch

The analytics added on 29 October measure the same things at scale: activation (pass a quiz within 24 h), Day-1 and Day-7 return, and the funnel sign-up → module → lesson → quiz → pass. Compare two weeks before and after each redesigned screen ships, and run Round 2 on the built app with 5 new participants.

8 Handoff and next steps

#	STEP	OWNER	EFFORT
1	Run Round 1 usability testing with the prototype; update screens	Product	1 week
2	Quick wins: zoom fix, Up Next fix, quiz focus mode, Logout → Me, pass/fail results	Dev	S (days)
3	Contrast fixes: muted text on grey, locked text, orange chips, field borders	Dev	S
4	Extract components: Sheet, Progress, PathNode, AnswerOption	Dev	M
5	Onboarding, explanations per answer, chapter path, lesson steps	Dev	M each
6	Review cards (spaced repetition) and daily goal	Dev	M–L
7	Module group chats with moderation (later stage)	Dev	L
8	Desktop layout (sidebar, side panel)	Dev	M

Files

FILE	WHAT IT IS
business/03-ux-ui-design.pdf	This document
business/prototype/index.html	Clickable prototype (15 screens, works offline)
business/design/screens/	High-fidelity PNGs (phone 2×, desktop 1.5×)
business/design/wireframes/	Wireframe PNGs
business/design/source/	HTML/Python source for every screen (edit and re-render)
docs/ux-review/	The 27 October UX review with current-app screenshots